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Palaeomagnetic and geochronological results from the Cambro-Ordovician Granite Harbour Intrusives inland of Terra Nova Bay (Victoria Land, Antarctica)

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SUMMARY

Palaeomagnetic investigations and Rb–Sr dating were carried out on samples from two plutons from the Granite Harbour Intrusives of the Transantarctic Mountains inland of Terra Nova Bay. The Rb–Sr whole rock–biotite ages from Teall Nunatak (475 ± 4, 483 ± 4 Ma), a quartz-diorite pluton cropping out to the south of Priestley Glacier, are older than that from the Mount Keinath monzogranite (450 ± 4 Ma), which is located to the north of the glacier. These results are consistent with the literature data, which suggest that during the last phases of the Ross Orogeny the cooling rate of the basement was significantly lower to the north than to the south of Priestley Glacier. The Teall Nunatak quartz-diorite is characterized by a stable magnetization, whose blocking-temperature spectrum ranges from 530 to 570 °C. At one site, the stable magnetization is screened by a large secondary component of opposite polarity, removed by thermal demagnetization below 300 °C. The characteristic directions after thermal demagnetization yielded a southern pole located at lat. 11°S, long. 21°E. The magnetization of Mount Keinath monzogranite consists of several components with overlapping stability spectra. A characteristic direction was isolated at one site only, obtained by demagnetizing the specimens in the temperature range from 380 to 460 °C.

Comparison with the other East Antarctica poles shows that those from Victoria Land are very well grouped and give a reliable early Ordovician palaeopole (lat. 5°S, long. 23°E, with $K = 196$ and $A_{95} = 3.7°$), whereas the poles from Wilkes, Enderby and Dronning Maud Land are dispersed. We tentatively advance the hypothesis that the dispersion reflects different magnetization ages due to the slow cooling of these regions during the last stages of the Ross Orogeny.

Key words: Antarctica, geochronology, Ordovician, palaeomagnetism, palaeopole.

INTRODUCTION

The Granite Harbour Intrusives (Gunn & Warren 1962) are syn- to post-tectonic granitoid plutons of Cambro-Ordovician age associated with the Ross Orogeny, which have intruded the Precambrian–Palaeozoic metamorphic basement of the Transantarctic Mountains. In northern Victoria Land, the Granite Harbour Intrusives occur as large, batholith-shaped, multiple intrusions of different lithologies. Three main magmatic suites have been recognized: high-K calc-alkaline, shoshonite and peraluminous granitoids (Borg et al. 1986; Ghezzo et al. 1989; Armienti et al. 1990a). The high-K calc-alkaline, metaluminous 'I' type association is the most widespread and is characterized by a wide compositional range (quartz-diorite, tonalite, granodiorite and monzogranite). The potassic syenogranite-monzonite association consists of several intrusive bodies, among which the Mount Abbot granite is the best known (Skinner 1983a; Ghezzo et al. 1989). Peraluminous granitoids were recognized by Borg et al. (1986, 1987), Vetter & Tessenshon (1987) and Ghezzo et al. (1989) and attributed to 'S' type association according to the Chappel & White (1974) classification. In northern Victoria Land, this association is restricted to the northern Deep Freeze Range and the Campbell–Aviator divide (Ghezzo et al. 1989).

In the region inland of Terra Nova Bay the Granite Harbour Intrusives have been grouped into three major batholiths, referred to as Campbell–Aviator (Nathan 1971), Terra Nova (Skinner 1983a; Skinner 1983b) and, to the south of Priestley

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Glacier, South Victoria batholith (Skinner 1983a, 1983b). The Terra Nova batholith is composed of a heterogeneous suite of plutons. Armienti et al. (1990b) further divided it in the large plutonic complex of Mount Abbot, cropping out south of Browning Pass, and the Deep Freeze Range Intrusives, between the Priestley and Campbell Glaciers (Fig. 1). The Mount Abbot rocks are characterized by significant K enrichment and mainly consist of syenogranite bodies intruded in small tonalite and granodiorite stocks, with minor occurrences of gabbroic and quartz-dioritic rocks. The composition of the Deep Freeze Range Intrusives ranges from diorite to monzogranite, with prevailing diorites and quartz-diorites (Armienti et al. 1990a).

The tectonic, metamorphic and magmatic events of the Ross Orogeny in central and northern Victoria Land are still not fully unravelled. Present geochronological data are sufficient only to outline the thermal history of the Granite Harbour Intrusives:

(1) peak-metamorphism conditions have been dated at about 540 Ma in the syn-tectonic, 'S'-type granitoids of the Wilson Plutonic Complex in the Rennick Glacier region (Adams 1986; Black & Sheraton 1990);

(2) the emplacement of the main suite of metaluminous granitoids occurred between 530 and 470 Ma (Tonarini & Rocchi 1994 and references therein), as shown by emplacement ages determined on single plutons by various authors (Borg et al. 1987 and references therein; Armienti et al. 1990b; Encarnacion & Grunow 1995);

(3) the cooling ages cluster around 460 Ma in the Deep Freeze Range and Mount Abbot granitoids and are higher than 480 Ma in the South Victoria batholith (Tonarini & Rocchi 1994 with references).

The ferromagnetic mineral content of the Ordovician granitoids of northern Victoria Land is usually low. The mean value of the magnetic susceptibility, measured on 259 samples, is 668×10^{-6} SI (Bozzo et al. 1992) and the remanence intensity is usually less than 1×10^{-4} A m^{-1} (Lanza, unpublished data). Two small plutons do not share these non-ideal characteristics: Teall Nunatak and Mount Keinath. Teall Nunatak belongs to the South Victoria batholith and is located (lat. 74.8°S, long. 162.5°E) at the mouth of Reeves Glacier (Fig. 1). It consists of granite and quartz-diorite. The quartz-diorite is composed of prevailing zoned plagioclase (bytownite-andesine) and associated ferro-horneblende and biotite, with subordinate quartz and sporadic microcline. Magnetite, apatite, titanite and zircon occur as accessory minerals. Preliminary sampling showed that the rock carries a stable remanence. 58 specimens were collected from four sites during the 1990–91 Italian expedition to Victoria Land. The Mount Keinath monzogranite pluton belongs to the Deep Freeze Range Intrusives. It crops out along the divide between the Priestley and Campbell Glaciers (lat. 74.5°S, long. 164.0°E), some 20 km to the north of Terra Nova Station. The rock is composed of zoned plagioclase (oscillating between andesine and oligoclase), quartz, microperthitic microcline, biotite and apatite, titanite, zircon and magnetite as accessories. The high ferromagnetic minerals content of this rock was first discovered from the regional magnetic map (Bozzo et al. 1989), which shows a large anomaly centred

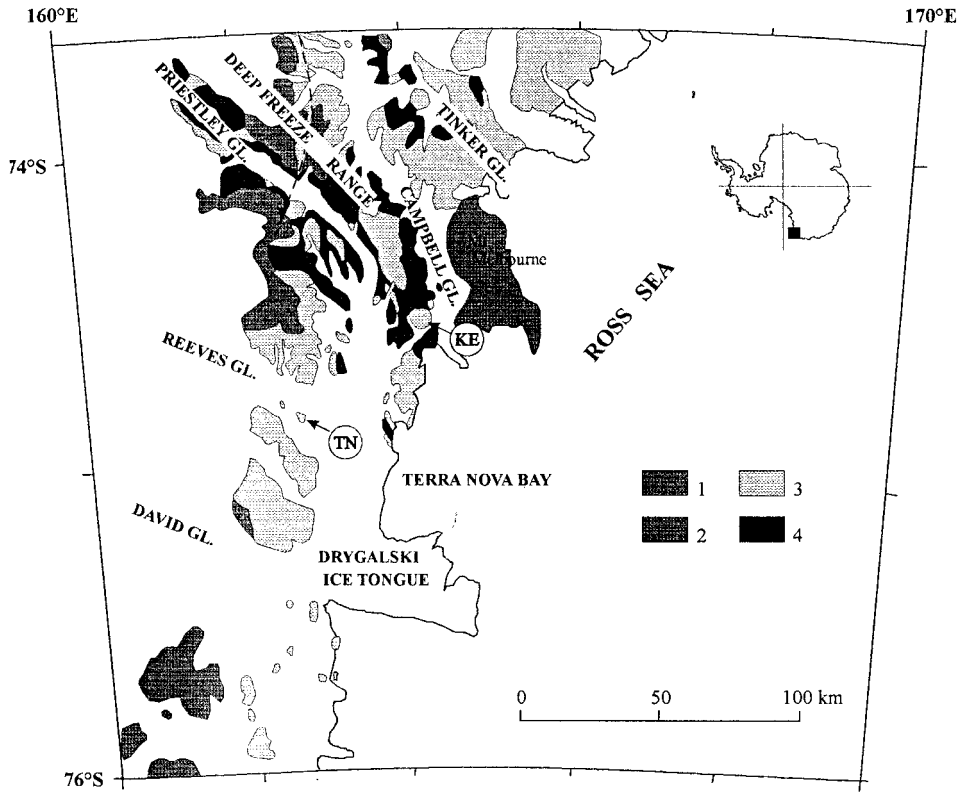


Figure 1. Geological sketch map (simplified from Carmignani et al. 1988). Symbols: (1) McMurdo Volcanics; (2) Ferrar dolerite and Kirkpatrick basalt; (3) Granite Harbour Intrusives; (4) Precambrian–Palaeozoic metamorphic basement. Sampling sites: TN = Teall Nunatak; KE = Mount Keinath.

on Mount Keinath. During the 1990–91 expedition, five sites were sampled for a total of 55 specimens.

The Teall Nunatak samples were oriented by both magnetic and solar compass. Bad weather conditions allowed only a few solar orientations at Mount Keinath, and the magnetic readings were corrected using the declination value measured at the nearby Terra Nova Geomagnetic Observatory (+137.5°).

GEOCHRONOLOGY

Geochronological data from the Granite Harbour Intrusives of northern Victoria Land reveal a long-lasting, complex structural evolution. In the region of this study, most radiometric data are mineral ages measured on plutonic and metamorphic rocks and should be interpreted as cooling ages. Rb–Sr whole rock–biotite ages show a regional distribution with values higher than 480 Ma in the South Victoria batholith, south of Priestley Glacier, and ranging from 470 to 445 Ma in the Deep Freeze Range and Mount Abbot Intrusives (Fig. 2). The only emplacement age, of 508±6 Ma, has been obtained from the Mount Abbot syenogranite and is some 50 Ma higher than the corresponding cooling ages determined on micas, which range from 450 to 443 Ma (Armienti et al. 1990b). The long time span between emplacement and cooling is thought to be the result of magma intrusion in a hot basement and an eventual slow cooling related to regional uplift. This hypothesis is also supported by the mineral ages determined on host migmatites, which are in the same range as those of the granitoids. A similar feature may be assumed for the Deep Freeze Range Intrusives, whose cooling age 'isochron' of 453±4 Ma (Armienti et al. 1990b) is close to that from Mount Abbot and suggests an almost synchronous closure of their Sr isotopic systems. The cooling age difference between the South Victoria batholith and the Deep Freeze Range Intrusives has been interpreted by Armienti et al. (1990b) as evidence for a major structural discontinuity separating two crustal sectors with different thermal and tectonic evolutions.

The regional distribution of radiometric ages requires caution in the correlation between geochronological and palaeomagnetic data. If the emplacement and cooling ages are similar, then both of them may be considered to correspond within the experimental error to the time of acquisition of the rock's primary magnetization. On the other hand, if the emplacement age is significantly older than the cooling age, then the Curie-point threshold (576 °C for magnetite) could have been passed much earlier than the closure temperature of 350 °C. As intrusive rocks whose magnetic carriers are dominated by multidomain grains may acquire their thermal remanent magnetization (TRM) over a large spectrum of blocking temperatures, a rock cooling over some tens of millions of years might record different palaeomagnetic directions, and its cooling age would correspond to the time of acquisition of the low-temperature, less stable components.

The samples investigated in this study were collected from the same outcrops as those used for palaeomagnetism and dated by the Rb–Sr whole rock–biotite method. The monzogranite from Mount Keinath yielded an age of 450±4 Ma (Table 1) that agrees well with the mineral ages reported for the Deep Freeze Range and fits the same biotite 'isochron' (Fig. 3). This result points to a cooling associated with regional uplift and suggests a slow cooling of the rock, which might therefore have acquired substantial secondary magnetization components. The samples from two different outcrops at Teall Nunatak gave ages of 475±4 and 483±4 Ma (Table 1), consistent with those from the South Victoria batholith. This intruded a basement consisting of low-grade metamorphosed sediments (Armienti et al. 1990b; Carmignani et al. 1988), which may be supposed to have been relatively cold at the time of intrusion. The cooling ages can thus be regarded as those of emplacement.

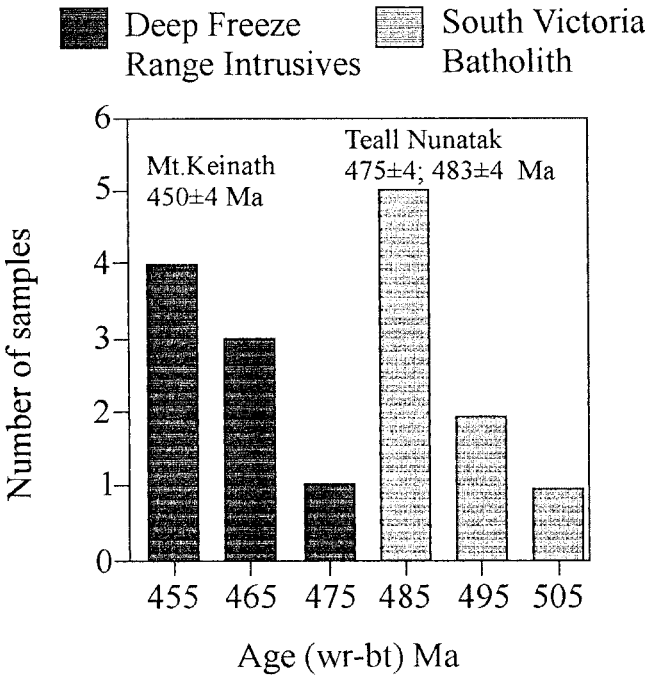


Figure 2. Rb–Sr whole rock–biotite ages of granitoids from the South Victoria batholith and Deep Freeze Range Intrusives.

PALAEOMAGNETISM

Magnetic susceptibility and natural remanent magnetization (NRM) of all specimens were measured with a KLY-2 bridge and a JR-4 spinner. Two pilot specimens for each site were stepwise demagnetized, one thermally and one in an alternating field; another specimen was used for measuring the susceptibility as a function of the temperature and a final one to investigate the isothermal remanent magnetization (IRM) up to a maximum field of 1.5 T.

Susceptibility versus temperature measurements, made using a Bartington MS2 instrument, showed Curie points around 550–570 °C (Fig. 4a). The results of thermal demagnetization were different for the two plutons, except that in both the NRM was completely removed at temperatures above 570 °C. The IRM measurements showed that saturation is always reached at field values between 0.1 and 0.2 T, and the remanent coercive force in the specimens from Mount Keinath (16–23 mT) is lower than in those from Teall Nunatak (50–70 mT, 28 mT at site TN3). These results point to low-Ti titanomagnetite as the main ferromagnetic mineral in both plutons, and suggest that the grains are mainly multidomain in the Mount Keinath monzogranite and (pseudo) single-domain in the Teall Nunatak quartz-diorite.

The NRMs of the two plutons are very different. At Teall Nunatak, a single-component NRM characterizes three of the

Table 1. Rb–Sr data and whole rock–biotite ages.

Sample	Rb ppm.	Sr ppm.	87Rb/86Sr	87Sr/86Sr ±2sigma	wr–bi age Ma±2sigma	(87Sr/86Sr)i
Mt. Keinath						
LZ 42 wr	136	274	1.44	0.71692±2		0.7077
biotite	580	4.69	461	3.6558±37	450±4	
Teall Nunatak						
LZ 27 wr	76.1	770	0.286	0.71124±4		0.7093
biotite	307	12	75.76	1.23019±6	483±4	
LZ 57 wr	67	741	0.261	0.71135±4		0.7096
biotite	418	13.76	93.11	1.33995±4	475±4	

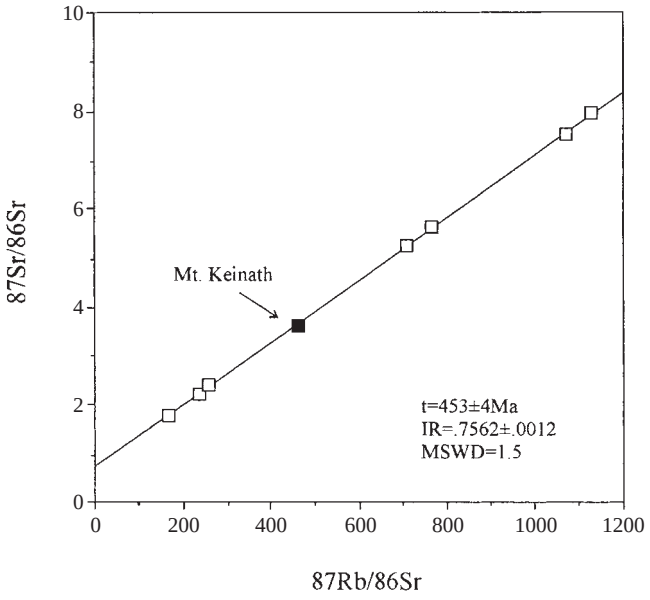


Figure 3. Cooling age 'isochron' of biotites from Deep Freeze Range Intrusives (modified from Armienti et al. 1990b).

four sites. It is very stable as its blocking-temperature spectrum ranges from 530 to 570 °C (Figs 4b and 5a). The fourth site, TN3, is characterized by a low remanent coercive force, very high susceptibility and low Konigsberger ration Q (Table 2). Its NRM directions are scattered, owing to the occurrence of large unstable magnetization components, which are completely removed by thermal demagnetization at temperatures below 300 °C (Figs 4b and 5b). At higher temperatures, a stable component opposite to the NRM is clearly isolated and its direction is the same as in the other three sites. The characteristic remanent magnetization (ChRM) at site TN3 is thus screened by a secondary component with the same direction but opposite polarity, and whose blocking temperatures are approximately 250–300 °C lower than the 530–570 °C that is typical of the ChRM. As we mentioned before, the South Victoria batholith is supposed to have cooled relatively quickly, and thus the age difference between the acquisition of the characteristic and the secondary magnetization should be small. We may reasonably conclude that the rock recorded a polarity inversion that occurred shortly after its emplacement. On the grounds of these results, all specimens from Teall Nunatak were thermally demagnetized at four steps in the range 530–570 °C, and the ChRM direction was computed by linear interpolation. The within-site dispersion of the directions was small, as shown in Fig. 6(a) and Table 2, which reports the mean-site results together with the values of the statistical parameters a_{95} and k.

The four mean-site directions were used to calculate the palaeomagnetic direction for the Teall Nunatak quartz-diorite: D=217.6°, I=−2.7°, with a_{95}=7.2° and k=163. The corresponding southern pole is located at lat. 11°S, long. 21°E.

The magnetization of the Mount Keinath monzogranite is rather different. The NRM directions are widely dispersed: at

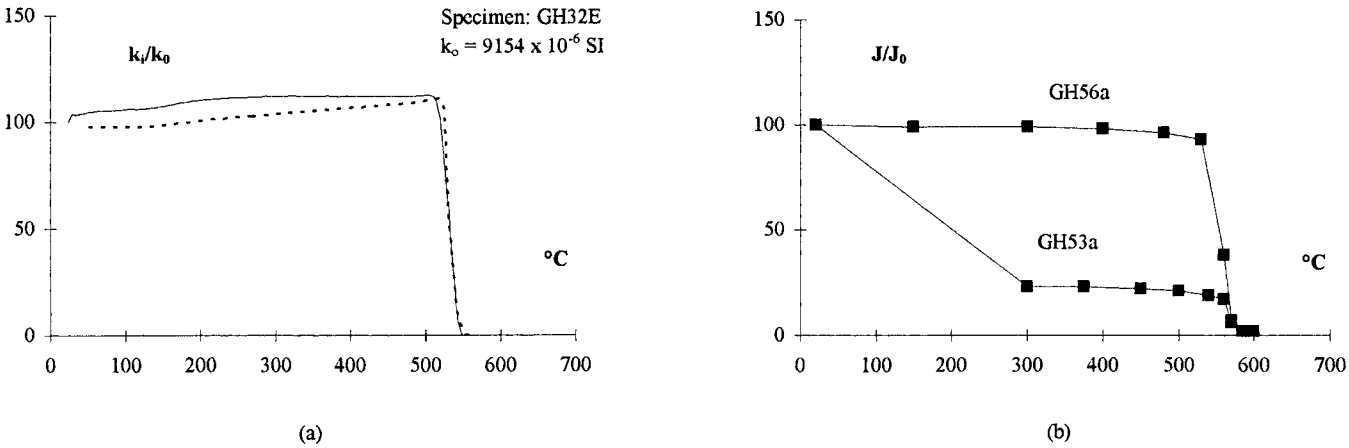


Figure 4. Magnetic properties versus temperature for the Teall Nunatak quartz-diorite. (a) Normalized susceptibility versus temperature. Full/dotted line: heating/cooling cycle; k_0: bulk susceptibility at room temperature. (b) Normalized intensity decay during thermal demagnetization for specimens GH56a (site TN4) and GH53a (site TN3).

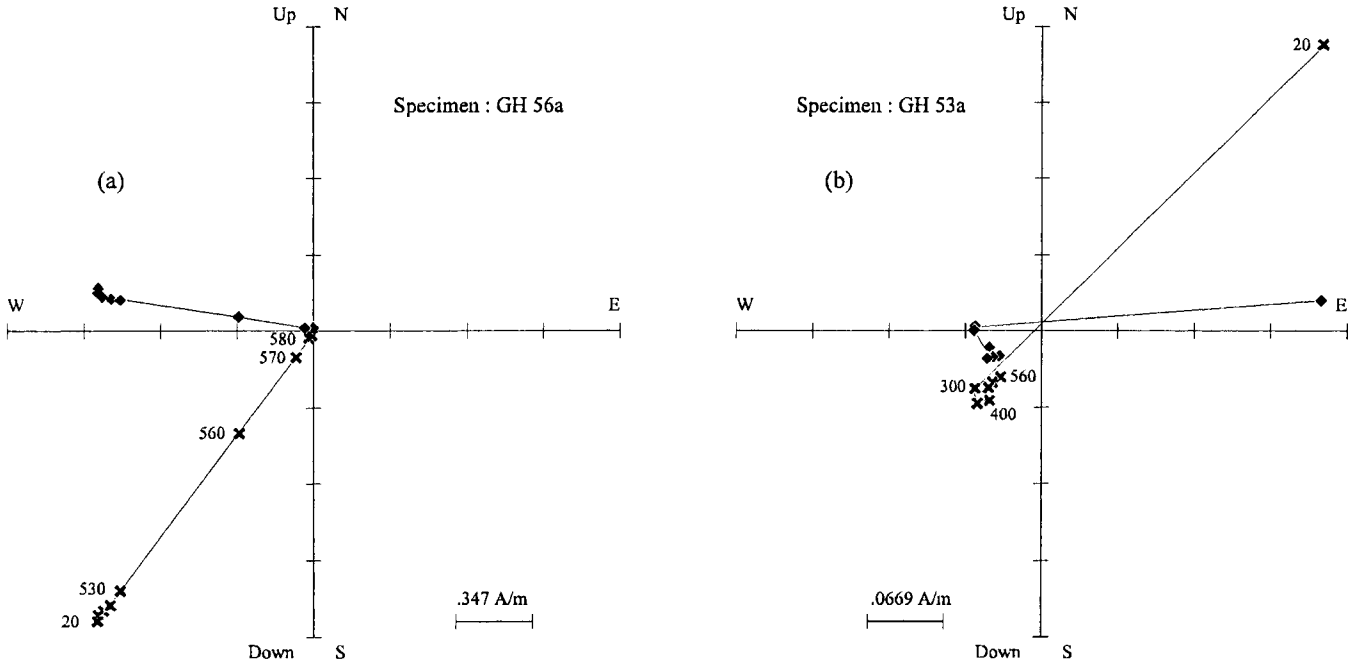


Figure 5. Zijderveld diagrams for thermal demagnetization (same specimens as in Fig. 4b). Diamonds: declination; crosses: apparent inclination; figures: temperature value in °C.

Table 2. Palaeomagnetic results from the Teall Nunatak (TN) and Mount Keinath (KE) plutons.

Site	N/n	k_m (SI × 10^6)	J_r (A m^{-1})	Q	NRM D	NRM I	NRM a_{95}	ChRM D	ChRM I	ChRM a_{95}	ChRM k
TN 1	12/12	891	0.01	0.2	199.9	+4.7	18.4	216.4	+3.3	5.4	67
TN 2	12/12	1898	0.12	1.2	219.7	−6.0	5.2	218.3	−5.1	3.0	208
TN 3	15/15	32 048	0.88	0.5		scattered		218.1	+1.3	5.3	53
TN 4	19/19	1539	1.53	19.1	215.8	−10.4	2.2	216.6	−10.3	2.5	181
All sites								217.6	−2.7	7.2	163
KE 1	7/9	15 876	0.06	0.07		scattered		219.7	+35.3	14.1	17

N/n: number of specimens collected/used in the calculation of the mean data; k_m: bulk susceptibility; NRM: natural remanent magnetization; J_r: NRM intensity; Q: Konigsberger ratio (calculated for a field intensity of 53 A m^{-1}); D, I: declination, inclination; a_{95}: semi-angle of confidence; ChRM: characteristic remanent magnetization; k: Fisher's precision parameter. Scattered: resultant vector R less than significance point R_0.

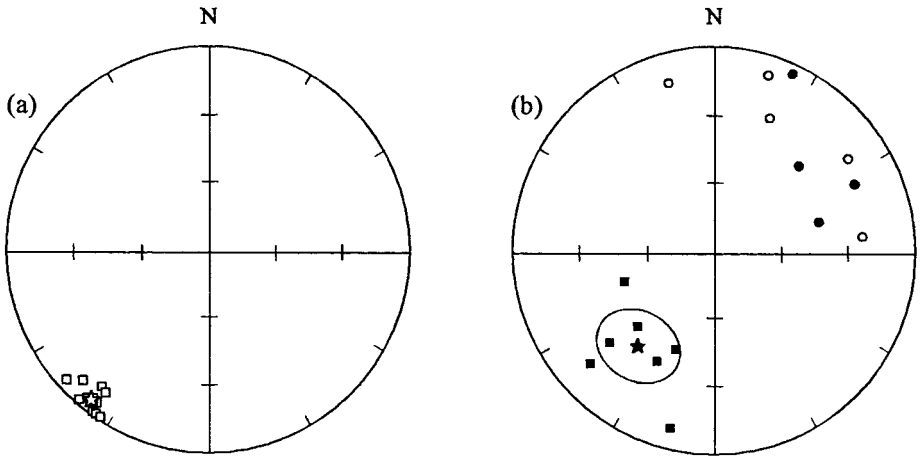


Figure 6. Equal-area projection of remanence directions: (a) Teall Nunatak, site TN4; (b) Mount Keinath, site KE1. Dots: NRM; squares: ChRM; stars: mean-site ChRM direction with a_{95} ellipse of confidence; full/open symbols: positive/negative inclination.

two sites the mean direction is characterized by an a_{95} value of approximately 30°, and at the other three sites the resultant vector R is less than the significance point R_0. Both thermal and alternating field demagnetizations usually failed to isolate a stable component. Thermal demagnetization was successful only at site KE1: the ChRM component of seven specimens was isolated by heating at several steps between 380 and 460 °C. The ChRM directions fall in the opposite quadrant with respect to the NRM directions, and are similar to those from Teall Nunatak, but with higher inclination (Fig. 6b). Also in this case, the characteristic and the secondary magnetizations share a similar direction and differ in polarity. The mean-site direction is D = 219.7°, I = +35.3°, with a_{95} = 14.1° and k = 17 (Table 2), which corresponds to a southern virtual geomagnetic pole (VGP) at lat. 31°S, long. 29°E.

DISCUSSION AND CONCLUSIONS

Geochronological data suggest that the different ages of the Teall Nunatak (475 ± 4 and 483 ± 4 Ma) and Mount Keinath (450 ± 4 Ma) plutons may be regarded as the outcome of different cooling histories. The Mount Keinath monzogranite stayed at higher temperatures for a longer time than the Teall Nunatak quartz-diorite, and its eventual cooling was mainly related to the regional uplift of the basement. This picture is consistent with the magnetization characteristics. A single-component NRM with a very high blocking-temperature spectrum is typical of the Teall Nunatak pluton, which gave a good palaeomagnetic direction and a reliable pole. On the other hand, the NRM of the Mount Keinath pluton consists of two or more components with overlapping blocking-temperature spectra; demagnetization was successful only for a few specimens, and the calculated VGP is tentative.

The secondary component discovered at Teall Nunatak at site TE3 has the same direction as the corresponding ChRM but with opposite polarity. Its very low inclination bears witness to an age only slightly younger than that of the ChRM. It also rules out the possibility of a magnetic overprint later than Ordovician, as the apparent polar wander (APW) path shows that East Antarctica left its equatorial position in late Ordovician (Thompson & Clark 1982) and did not occupy it again. On laboratory timescales, the blocking temperature of the secondary magnetization is lower than 300 °C. Transposing this value to geological times (Pullaiah et al. 1975; Dunlop 1983) means that reheating at temperatures as low as 200–300 °C would have been enough to unblock the secondary magnetization and result in a magnetic overprint during the eventual cooling. Teall Nunatak was therefore affected by neither the intrusion of the Jurassic Ferrar dolerite sills, which are known to have overprinted the basement rocks in various sites of the Transantarctic Mountains (Bull et al. 1962; Beck & Schmidt 1971; Funaki 1984; Grunow & Encarnacion 1993), nor the Cretaceous thermal event suggested for the Bowers Supergroup rocks of northern Victoria Land (Schmierer & Burmester 1986).

Table 3 and Fig. 7 report all the Ordovician poles for Antarctica, with the exception of those derived from only one site which largely deviate from the others (Tokieda & Funaki 1991). The positions of three poles differ from those reported in the original papers:

(1) the Mirnyy Station pole (McQueen et al. 1972) has been recalculated leaving out the sites with $a_{95} > 30°$;

(2) the Ongul Islands pole (Funaki & Wasilewski 1986) has been recalculated using the correct value of the WON2 VGP (lat. 35.6°S, long. 13.7°E, Funaki, personal communication) and leaving out the site with $a_{95} = 37°$;

(3) the Sør Rondane dolerite pole (Tokieda & Funaki 1991) has been calculated from the mean of the three investigated sites.

Table 3 also lists the rock ages as reported in the original papers. To assign a definite age to the thermal remanent magnetization of a rock requires knowledge of both the rock's thermal history and blocking-temperature spectrum. The temperature range used to derive the ChRM (and thus the VGP) throughout the demagnetization process may well differ from the closing temperature of the isotopic system used for dating. When we deal with intrusive and metamorphic rocks, the cooling rate is low and the blocking-temperature spectrum is often wide: the magnetization and radiometric ages do not necessarily coincide.

The ChRM directions used to calculate the Victoria Land VGPs were all obtained after thermal demagnetization, the only exception being those from the lamprophyres from Taylor Valley (Manzoni & Nanni 1977), which were demagnetized in alternating fields. Most VGPs have been derived from a temperature range of 500–570 °C, higher than the closing temperature of Rb–Sr and K–Ar biotite systems (300–350 °C, Blanckenburg et al. 1989). The magnetization age may therefore be assumed to be slightly older than the mineral ages, which are close to the emplacement ages (Tonarini & Rocchi 1994; Grunow 1995). Leaving aside our VGP from Mount Keinath, the poles from Victoria Land are mutually consistent and we may confidently regard their mean location (lat. 5°S, long. 23°E with K = 196, A_{95} = 3.7°) as the early Ordovician (≈490–480 Ma) palaeopole. We have so far left out the possibility that tilt movements may have occurred after the ChRM acquisition and biased directions. Grunow (1995) has already noted that the consistency of data from Victoria Land argues against local tilting in the different sampling regions. A regional tilt, on the other hand, could have occurred, since the Beacon Group sediments, which unconformably overlie the basement and the Granite Harbour Intrusives, dip gently (2°–5°) towards the west at the regional scale. Such a small westerly tilting would have had very limited effects on the ChRM directions, which point towards S–SSW with low inclination. The corresponding poles would move some 2° northwards.

The interpretation of the results from other regions of East Antarctica (Wilkes, Enderby and Dronning Maud Land) is more difficult. The poles are rather dispersed: two (Mirnyy Station and Mawson Station) fall near the Victoria Land poles, the others are displaced some 10° to 25° towards higher southern latitudes (Fig. 7). Most of the rocks studied were demagnetized in alternating fields, and information concerning the stability of the magnetization with increasing temperature is only available for the Larsemann Hills, Scullin Monolith and Mawson Station VGPs, derived from ChRM directions after demagnetization at 380 °C, 380 °C and 450 °C, respectively (Funaki & Saito 1992). The lack of data and the complex thermal history of the regions make it difficult to constrain the timing of the magnetization. In Dronning Maud Land, a cooling rate significantly lower than in the Transantarctic

Table 3. Ordovician poles from East Antarctica.

Locality	Lithotype	Age (Ma)	N/n	Palaeo-latitude	Pole Lat. S	Pole Long. E	Reference
Victoria Land							
1 Taylor Valley	Lamprophyre	~470	**4**/51	0°	9°	27°	Manzoni & Nanni (1977)
2 Wright Valley	Lamprophyre	470±7	?/**58**	7°N	2.5°	24°	Funaki (1984)
3 Wright Valley	Granite	500±43	?/**16**	5°N	5°	18.5°	Funaki (1984)
4 Granite Harbour	Grey granite	476±15	**5**/34	6°N	4°	22°	Grunow (1995)
5 Granite Harbour	Pink granite	478±35	3/**21**	8°N	3°	17°	Grunow (1995)
6 Lake Vanda, Briggs Hill, Granite Harbour	Mafic dikes	?	**9**/48	9°N	2°	18°	Grunow (1995)
7 Lake Vanda, Granite Harbour	Feldspar porphyry	467±15	**8**/41	5°N	3°	28°	Grunow (1995)
8 Lake Vanda, Briggs Hill	Granodiorite	472±15	**5**/36	2°N	6°	31°	Grunow (1995)
9 Teall Nunatak	Quartz-diorite	475±4, 483±4	**4**/58	1°N	11°	21°	This paper
10 Mount Keinath	Monzogranite	450±4	1/**7**	19°S	31°	29°	This paper
East Antarctica craton							
11 Mirnyy Station	Charnockite	502±4	**2**/18	11°S	3°	25°	McQueen et al. (1972)*
12 Larsemann Hills	Biotite gneiss	444±3	?/**23**	44°S	33°	28°	Funaki & Saito (1992)
13 Scullin Monolith	Granitic gneiss	450±5	?/**5**	30°S	26.5°	353	Funaki & Saito (1992)
14 Mawson Station	Charnockite	445±27	?/**88**	25°S	9°	23°	Funaki & Saito (1992)
15 Syowa Station	Granitic gneiss	483±9	?/18	39°S	19°	17°	Nagata & Shimizu (1959)
16 Ongul Islands	Intrusives, gneiss	~480	**4**/75	40°S	24°	19°	Funaki & Wasilewski (1986)*
17 Sør Rondane Mountains	Dolerite	439±13	**3**/25	45°S	29°	6°	Tokieda & Funaki (1991)*
18 Sør Rondane Mountains	Syenite	496±9, 500±9	1/**9**	44°S	29.5°	353.5°	Tokieda & Funaki (1991)
19 Sør Rondane Mountains	Intrusives	~490	3/16	46°S	28.5°	9.5°	Zijderveld (1968)

N/n: number of sites/specimens (the bold character indicates which directions, either sites or individual specimens, have been used in the calculation of the mean direction for the locality); *: pole recalculated from the original data (see text for more explanation).

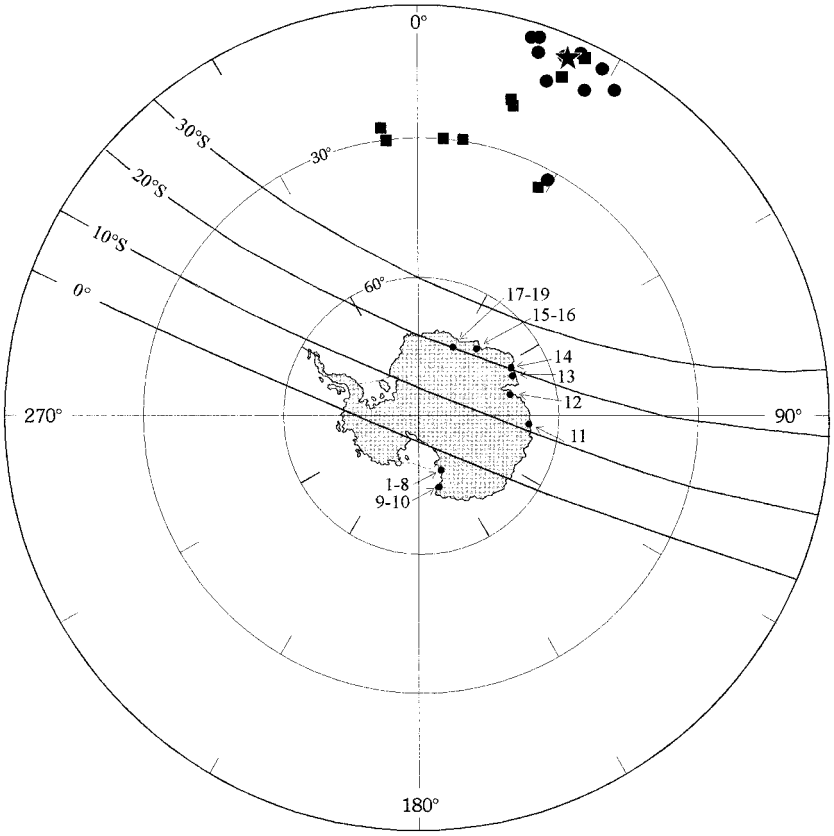


Figure 7. Equal-area projection of the Ordovician poles from East Antarctica. Dots: Victoria Land; squares: Wilkes, Enderby and Dronning Maud Land; small dots: samplig localities (figures refer to Table 3). The early Ordovician southern palaeopole from Victoria Land (star) and the palaeoparallels 0° to 30°S are also shown.

Mountains is well documented (Moyes, Barton & Groenewald 1993), and slow cooling has also been suggested for the Sør Rondane Mountains (Takigami & Funaki 1991). It is worth noting that the Larsemann Hills and Scullin Monolith VGPs, whose latitude is about 30°S, were obtained after demagnetization at the relatively low temperature of 380 °C. The Larsemann Hills VGP is very close to the Mount Keinath VGP, which was obtained in the temperature range of 380–460 °C and whose magnetization is at least 30 Myr younger than that of other localities in Victoria Land. Assuming that a lower temperature for the ChRM acquisition matches a lower cooling rate of the rock, the poles displaced southwards could be younger than early Ordovician and reflect apparent polar wandering. At the moment, this hypothesis is only tentative and needs to be substantiated by more data on the thermal characteristics of the magnetic remanence and the thermal history of the various sampling regions. It is, however, consistent with:

(1) the palaeolatitudes of the Mount Keinath, Larsemann Hills, Ongul Islands and Sør Rondane Mountains sites (Table 3, Fig. 7), which are some 20°–30° higher than the parallels corresponding to the Victoria Land early Ordovician palaeopole; and

(2) the synthetic East Antarctica APW path (Thompson & Clark 1982), which shows that in the early Ordovician the palaeopole started to move towards the South Geographic Pole and that this motion lasted throughout the Palaeozoic.

Comparison with poles from other landmasses of Gondwana is of little assistance, since the Ordovician segment of their APW path is poorly defined (Schmidt et al. 1990). The available data are few and most suffer from the same uncertainties as we have seen for Antarctica: poor age assignment, lack of chronological correlation between the cooling and remanence acquisition processes, and later magnetic overprints. To illustrate the intricacies of the matter, we refer to Australia, which was the landmass adjacent to East Antarctica. After reassembling the two continents, the pole from the Jinduckin Formation in the Australian Northern Territory (Luck 1972) is the only Ordovician pole fully consistent with the results from South Victoria Land. On the other hand, the angular distance between our pole from Teall Nunatak (475 ± 4, 483 ± 4 Ma) and that from the Black Hill Norite (487 ± 5 Ma) in the Adelaide Geosyncline of South Australia (Schmidt et al. 1993) is about 15°, even though the pole ages are quite similar. A much greater distance lies between the South Victoria Land poles and those of Cambro-Ordovician age derived from rocks overprinted during the Delamerian Orogeny in various Australian localities (Klootwijk 1980). It is worth noting that the results of the reassembling exercise do not substantially depend on the rotation poles used (Smith & Hallam 1970; Besse & Courtillot 1988; Royer & Sandwell 1989).

This study has contributed a new pole to the East Antarctica APW path and pointed to the need for more reliable timing of palaeopoles in order to define its Ordovician segment better.

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